HSIN-PEI CHEN

Ph.D. Student in Astronomy, The University of Texas at Austin

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RESEARCH INTERESTS

Computational astrophysics; star formation and astrochemistry; cosmic-ray ionization; stellar evolution; binary interactions and supernova progenitors.

EDUCATION

The University of Texas at Austin (UT Austin) Aug. 2024–Present
Ph.D. Program in Astronomy Austin, TX, USA
Advisor: Dr. Stella S. R. Offner
Second-Year Assessment: **Full Pass** (Spring 2026); eligible to apply for Ph.D. candidacy.

National Tsing Hua University (NTHU) Sep. 2020–Jan. 2023
Master of Science in Astronomy Hsinchu, Taiwan
Advisor: Dr. Kuo-Chuan Pan

National Kaohsiung Normal University (NKNU) Sep. 2016–Jun. 2020
Bachelor of Science in Physics Kaohsiung, Taiwan

RESEARCH & TEACHING APPOINTMENTS

Graduate Research Assistant, Department of Astronomy, UT Austin — Spring 2026–Present
Teaching Assistant, Stellar Astronomy (AST 352K), UT Austin, Prof. Harriet Dinerstein — Fall 2025
Graduate Research Assistant, Department of Astronomy, UT Austin — Fall 2024–Summer 2025
Research Assistant, Institute of Astronomy, NTHU — Dec. 2023–May 2024
Research Assistant, Center for Theory and Computation, NTHU — Mar. 2023–Nov. 2023
Teaching Assistant, Introduction to Astrophysics, NTHU, Prof. Daniel Harsono — Fall 2021
Teaching Assistant, Introduction to Astronomy, NKNU, Prof. Chien-Wen Hwang — Fall 2018

RESEARCH EXPERIENCE

Probing Cosmic-Ray Ionization Rates in Star-Forming Clouds: A Synthetic Observation with the STARFORGE Simulations Aug. 2024–Present
Developing synthetic observations of STARFORGE star-forming clouds with UCLCHEM to evaluate how reliably molecular tracers recover cosmic-ray ionization rates and investigate discrepancies between direct and indirect measurements.

Type Ia Supernova Progenitors and Surviving Companions within the Symbiotic Channel Feb. 2023–Jul. 2025
Modeled symbiotic Type Ia supernova progenitors with MESA across four classes of red-giant and AGB companions; predicted either a faint blue dwarf or a companion resembling its pre-supernova state as the surviving companion.

Exploring the Observability of Surviving Companions of Stripped-Envelope Supernovae: A Case Study of Type Ic SN 2020oi May 2021–Jan. 2023
Simulated ejecta–companion interaction and post-impact stellar evolution for SN 2020oi using FLASH and MESA; predicted that a main-sequence surviving companion may become detectable within a decade.

PUBLICATIONS

[1] Yu-Hui Wang, **Hsin-Pei Chen**, and Kuo-Chuan Pan, "Type Ia Supernova Progenitors and Surviving Companions within the Symbiotic Channel," *The Astrophysical Journal*, **989**, 72 (2025).
DOI: 10.3847/1538-4357/adeb71**.** ***Co-first author.***

[2] **Hsin-Pei Chen**, Shiau-Jie Rau, and Kuo-Chuan Pan, "Exploring the Observability of Surviving Companions of Stripped-Envelope Supernovae: A Case Study of Type Ic SN 2020oi," *The Astrophysical Journal*, **949**, 121 (2023).
DOI: 10.3847/1538-4357/acc9af**.**

HONORS

2024 Wu Chien-Shiung Scholarship (Master's Student/Graduate), The Physics Society of Taiwan.

ORAL PRESENTATIONS

Research Talk, "Measuring Cosmic Ray Ionization Rates in the STARFORGE Simulations: How well do H_3^+ observations measure CR ionization?," *2026 Fall Stars, Planets, and ISM Seminar at UT Austin* Oct. 2026

Contributed talk, "Measuring Cosmic Ray Ionization Rates in the STARFORGE Simulations: How well do H_3^+ observations measure CR ionization?," *Cosmic Rays 4: the Salt of the Star Formation Recipe*, Pisa, Italy Oct. 2026

Second-Year Assessment Research Presentation, "Cosmic Ray Ionization Rates in Star-Forming Clouds from STARFORGE Simulations Compared with Observations," Dept. of Astronomy, UT Austin Mar. 2026

Contributed talk, "Probing Cosmic Ray Ionization Rates in Star-Forming Clouds with the STARFORGE Simulations," *2025 Star and Planet Formation in the Southwest Conference* Dec. 2025

Research Talk, "Exploring the Observability of Surviving Companions of Stripped-Envelope Supernovae: A Case Study of Type Ic SN 2020oi," *2024 Fall Stars, Planets, and ISM Seminar at UT Austin* Oct. 2024

Contributed talk, "Exploring the Observability of Surviving Companions of Stripped-Envelope Supernovae: A Case Study of Type Ic SN 2020oi," *2023 ASROC Annual Meeting* May 2023

Contributed talk, "Exploring the Observability of Surviving Companions of Stripped-Envelope Supernovae: A Case Study of Type Ic SN 2020oi," *9th Eastern Asian Numerical Astrophysics Meeting* Sep. 2022

POSTERS

2025 Texas Advanced Computing Center Symposium Sep. 2025
CraigFest 2024: Celebrating the Life & Career of J. Craig Wheeler Oct. 2024
Poster Competition of the Physics Department at NTHU Dec. 2022
2022 Physical Society of Taiwan Annual Meeting Jan. 2022

SKILLS

Programming Languages	Python, Fortran, MATLAB
Scientific Software	Gizmo, UCLCHEM, Slabspec, FLASH, MESA, CASA
Scientific Computing	Linux, Slurm, high-performance computing
Development Tools	Git, GitHub, LaTeX
Scientific Machine Learning	PINNs, neural ODEs, operator learning, and automatic differentiation (graduate coursework)

SERVICE & OUTREACH

President, Taiwanese Student Association at UT Austin	May 2026–Present
Activity Coordinator, Taiwanese Student Association at UT Austin	Aug. 2025–Apr. 2026
Director, Astronomy Club Union of Universities in Taiwan (ACUUT)	Mar. 2021–Mar. 2023
General Coordinator, Taiwan Solar Eclipse Live-Streaming with Central Weather Bureau	Jun. 2020
President and Co-founder, Astronomy Club at NKNU	Jul. 2017–Jun. 2018